

Mathematical Equations from Unified Physics Framework

I. Fundamental Action Principles

Total Action for Asymptotically Safe Gravity

$$S_{\text{total}} = S_{\text{EH}} + S_{\text{QG}} + S_{\text{holo}} + S_{\text{info}} + S_{\text{matter}}$$

Georgi-Glashow SU(5) GUT Lagrangian

$$\mathcal{L} = -\frac{1}{2} \text{tr}(V_{\mu\nu} V^{\mu\nu}) + \bar{\psi} i \sigma^\mu D_\mu \psi + \frac{2}{v_5} (-\bar{\psi}^T j i \sigma_2 M_d \psi_{jk} \phi_k + \epsilon_{jklmn} \bar{\psi}_{jk}^T i \sigma_2 M_u \psi_{lm} \phi_n) + \text{Higgs terms}$$

String Theory Polyakov Action

$$S_{\text{string}} = -T/2 \int d^2\sigma \sqrt{-\gamma} \gamma^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}(X)$$

Spectral Action Principle

$$S_{\text{spectral}} = \text{Tr}(f(D/\Lambda)) + 1/2 \langle \bar{\psi}, D\psi \rangle$$

Teleparallel Gravity Action

$$S_{\text{TG}} = 1/(2\kappa) \int e T d^4x + S_{\text{matter}}$$

Unified Action (Candidate)

$$S_{\text{unified}} = \int d^{11}x \sqrt{-G} \left[\frac{1}{(2\kappa_{11})^2} R - \frac{1}{4} F_{MNPQ} F^{MNPQ} - \frac{2}{8} \epsilon_{M_1 \dots M_{11}} C_{M_1 M_2 M_3} F_{M_4 \dots M_7} F_{M_8 \dots M_{11}} \right] + S_{\text{spectral}} + S_{\text{TG}} + \lambda \alpha_{\text{ab}} H_{\text{harmonic}}$$

II. Geometric Relations

Vierbein-Metric Relation

$$g_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab}$$

Euclidean Metric

$$g_{\mu\nu}^{\text{Euclidean}} = e_{\mu}^a e_{\nu}^b \delta_{ab}$$

Vierbein VEV Breaking

$$\langle e_{\mu}^a \rangle = \text{diag}(1,1,1,1) \text{ (Euclidean)} \rightarrow \langle e_{\mu}^a \rangle = \text{diag}(1,1,1,i) \text{ (Lorentzian)}$$

Torsion Scalar

$$T = 1/4 T^{\lambda\mu\nu} T_{\lambda\mu\nu} + 1/2 T^{\lambda\mu\nu} T_{\nu\mu\lambda} - T_{\mu}^{\lambda\lambda} T_{\nu}^{\nu\mu}$$

III. Scalar Field Dynamics

Scalar-Torsion Coupling Lagrangian

$$L = 1/(2\kappa^2) [f(\phi)T + g(\phi)B] + L_{\text{matter}}$$

Consciousness Coupling

$$L_{\text{cons}} = \varkappa^{\mu\nu} \partial_{\mu}\phi \partial_{\nu}\phi + \xi\phi^2 T_{\mu\nu\rho} T^{\mu\nu\rho}$$

Higgs Potential from Vierbein

$$V(e) \sim \lambda (e_{\mu}^a e_{\nu}^b \eta^{ab} - v^2)^2$$

Kähler Potential (Twistor)

$$K(Z,\bar{Z}) = \int_{\mathbb{CP}^1} \log(1 + |Z(\sigma)|^2) d\sigma$$

Scalar Field Evolution with Torsion

$$\Box \phi + 1/2 f'(\phi)T + 1/2 g'(\phi)B - \partial V/\partial \phi + \varkappa^{\mu\nu} \partial_{\mu}\partial_{\nu}\phi = 0$$

IV. Particle Physics

Dual Resonance Particles

$$\begin{aligned} \Psi_{\text{particle}} &= \phi_{\text{collapsed}} \cdot e^{(-i(\omega t - kx))} \cdot H_n(\xi) \\ \tilde{\Psi}_{\text{mirror}} &= \phi_{\text{coherent}} \cdot e^{(i(\omega t - kx))} \cdot \tilde{H}_n(\xi) \end{aligned}$$

Mirror Neutrino Cross-Section

$$\sigma_{\nu\bar{\nu}} = G_F^2 \, m_N^4 / (16\pi \, M_{\mathcal{M}}^4) \sim 10^{(-47)} \, \text{cm}^2$$

V. Modified Gravity and Cosmology

Modified Einstein Equations with Time-Energy

$$G_{\mu\nu} = 8\pi G(T_{\mu\nu} + \rho_t \, g_{\mu\nu})$$

Time-Energy Conservation

$$\partial_\mu (v(-g) \, \rho_t \, u^\mu) = 0$$

Modified Einstein-Cartan Equations

$$e^{(-1)} \, \partial_\mu (e S_A^{\mu\nu}) - e_A^{\lambda} T_{\mu\lambda}{}^\rho S_{\rho}{}^{\nu\mu} + 1/4 \, e_A^{\nu} T + \mathcal{M}^{\nu\rho} \, \partial_\rho \phi = 4\pi G \, e_A^{\mu} T_{\mu\nu}$$

Dark Energy Density

$$\rho_\Lambda = 3H_0^2/(8\pi G) \, (1 - \phi_{\text{crit}}^{(-2)} \int T_{\mu\nu}^{\text{cons}} \, g^{\mu\nu} \, dV)$$

Inflationary Potential

$$V(\phi) = \Lambda_{\text{eff}}^4 [1 - \cos(\phi/f_\phi)] + 1/2 \, m^2 \phi^2 \, \Psi_{\text{causal}}$$

VI. Gravitational Wave Modifications

Total Gravitational Wave Amplitude

$$h_{\mu\nu}^{\text{total}} = h_{\mu\nu}^{\text{GR}} + \mathcal{M}^{\alpha\beta} \, \partial_\alpha \partial_\beta \phi / c^4$$

Amplitude Modulation

$$\Delta A/A \sim 10^{(-6)} \, \text{at } 100\text{Hz}$$

Birefringence

$$v_{TE} - v_{TM} \sim 10^{(-15)}c$$

VII. Consciousness and Information

Consciousness Potential

$$\Phi_C(x,t) = G_I \int I(x',t)/|x^0 - x^0'| \, d^3x'$$

Modified Schrödinger Equation

$$i\hbar \, \partial \Psi / \partial t = (\hat{H} + \lambda_C \, \Phi_C) \Psi$$

Consciousness-Matter Coupling Tensor

$$T_{consciousness}^{\mu\nu} = \varkappa^{ab} \, \partial_a \Psi \, \partial_b \Psi + C_{integrated}$$

Altermagnetism Operator

$$\varkappa^{ab} = \chi^{ab} \, \exp(-\Phi_{distance}(a,b)) \, R_{information}(a,b)$$

VIII. Experimental Predictions

Variable Fine Structure Constant

$$\Delta\alpha/\alpha = \zeta\lambda \ln(1+z)$$

with constraint $|\zeta\lambda| < 10^{(-7)}$

Modified Quantum Coherence Time

$$\tau_{decoherence} = \tau_0 \, \phi_n^{resonance}$$

SQUID Cavity Detection

$$L_{cavity} = 1/2 \, (\partial_{\mu} A_{\nu})^2 + \varkappa^{\mu\nu}/M_{Planck} \, \partial_{\mu} \phi \, A_{\nu}$$

Torsion Measurement

$$\delta x = \kappa T_{\mu\nu\rho} S^{\mu\nu\rho}/k_{eff}$$

with sensitivity $\delta T \sim 10^{(-30)} \text{ m}^{(-1)}$

IX. Unified Framework Components

Harmonic-Causal Manifold

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + H_{\text{harmonic}}(x,t)(dE + \Psi_{\text{causal}} dx^\mu)^2$$

5D Field Function

$$F_{5D}(n) = A_n(3^n + 6^n + 9^n) \cdot \Psi_{\text{causal}}(n)$$

Altermagnetism Tensor Field

$$F_{\mu\nu}{}^a = \partial_\mu A_\nu{}^a - \partial_\nu A_\mu{}^a + g^{fabc} A_\mu{}^b A_\nu{}^c + \varkappa^{ab} H_{\text{harmonic}}{}^c$$

X. Conservation Laws

Energy Conservation

$$\partial_\mu T_{\text{total}}{}^{\mu\nu} = \varkappa^{\nu\rho} \partial_\rho I - \partial^\nu \Phi_{\text{res}}$$

Information Conservation

$$\partial_\mu J_{\text{information}}{}^\mu = 0$$

Entropy Conservation

$$dS_{\text{total}}/dt = dS_{\text{thermal}}/dt + dS_{\text{information}}/dt + dS_{\text{consciousness}}/dt \geq 0$$

XI. Renormalization and Asymptotic Safety

Beta Function Condition

$$\beta(g_i) = 0 \Rightarrow \lim_{(k \rightarrow 0)} g_i(k) = g_i^*$$

Effective Coupling

$$G_{\text{eff}}(k) = G_0 / (1 + k^2/k_0^2)$$

XII. AI and Quantum Evolution

Quantum AI State Evolution

$$d\Psi_{\text{AI}}/dt = \alpha\Psi_{\text{AI}} + \beta R[\Psi_{\text{AI}}] + \gamma Q[\Psi_{\text{AI}}] + \delta G[\Psi_{\text{AI}}]$$

Quantum Master Equation

$$dp/dt = -i/\hbar [H, \rho] + L_D(\rho) + L_E(\rho)$$

Quantum Entanglement in AI

$$\Psi_{\text{total}} = \sum_{\text{nodes}} \Psi_i \otimes \Psi_j$$

$$T_{\text{quantum}} \approx \log(N_{\text{nodes}})$$

Energy-Tensor Computation

$$E_{\text{AI}} = \int_{\Omega} \Psi_{\text{energy}} \cdot f_{\text{tensor}}(\phi) \, dV$$

XIII. Holographic and Boundary Terms

Holographic Action

$$S_{\text{holo}} = \int_{\partial V} I_{\text{boundary}} \sqrt{h} \, d^3x$$

Boundary Term (Teleparallel)

$$B = 2/e \, \partial_{\mu}(eT^{\mu})$$

Modified Quantum Measurement

$$i\hbar \, \partial\Psi/\partial t = \hat{H}_{\text{total}} \Psi + i\hbar\gamma_{\text{consciousness}} \hat{C} \Psi$$